

From Propositional to Perceptual Asymmetry

Extending Frictive Policy Optimization to Asymmetric Partial Information Dialogue

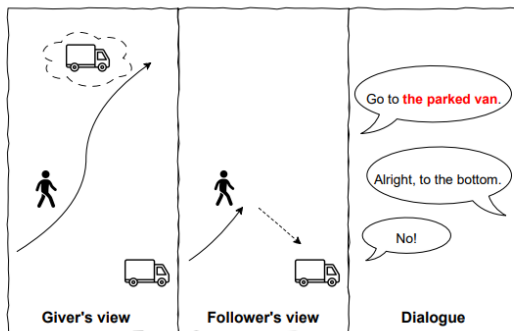
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This is a communication breakdown

PROBLEM



The giver's map has **two** parked vans; only one is intended. The follower's map has **just one** — and it's the wrong one. "The parked van" is a perfectly formed instruction. It still breaks.

Why can't an agent catch this?

PROBLEM

Frictive Policy Optimization (FPO; Pustejovsky & Krishnaswamy 2025, 2026) trains agents to treat misalignment as a signal worth intervening on, not noise to suppress:

$$F(h, y) = F^+(h, y) - F^-(h, y)$$

F^+ : productive friction (info gain). F^- : epistemic risk.

But every implementation assumes both participants read the same history h .

The gap

When $h^{(g)} \neq h^{(f)}$, there is no shared history left to evaluate friction over.

Propositional asymmetry

Same scene, different beliefs. **Covered.**

Perceptual asymmetry

Different evidence entirely, like the two maps you just saw. **Not covered.**

OUR APPROACH

Extend FPO so friction is evaluated relative to each participant's own information horizon $h^{(i)}$ — never a shared frame. Three things follow:

1. **Show the problem is real and structured** (MapTask corpus): perceptual asymmetry isn't rare noise — it's concentrated in a specific, identifiable pattern.
2. **Build the fix**: a perspective-bound friction taxonomy, answerable from one participant's view alone.
3. **Prove the fix works**: perspective-restricted LLMs detect misalignment *better* than omniscient ones — and the signal predicts real repair.

Data: HCRC MapTask, two annotation layers

EVIDENCE

Task: *giver* holds a map with a route; *follower* holds a *different* map of the same area — exactly the setup you just saw.

NXT layer (Anderson et al., 1991)

Dialogue-act moves: instruct, check, clarify,
...

Grounded Misunderstandings in MapTask dataset (GMMT) layer (Li et al. 2025)

13,077 REs, 128 dialogues. Per RE: *giver's intended* vs. *follower's interpreted* landmark $\Rightarrow$
aligned / **pending** / **misunderstood**

Joined at (dialogue_id, utterance_id) — no re-annotation needed.

Where misunderstanding concentrates

EVIDENCE

Discrepancy	% of REs	Misund. rate
Identical	63.7%	0.2%
Lexical	7.0%	1.4%
Multiplicity	7.3%	12.0%
Existence	22.0%	3.2%

6×

the corpus-average
misunderstanding rate,
concentrated in just
7% of landmarks

This is exactly the failure you saw in the opening example — and it's not an edge case, it's a structural pattern we can point an agent at.

Our fix: perspectival frictive states

SOLUTION

Each participant can spot friction using only their own view — never the partner's. A frictive state is now evaluated *per participant*: $F^-(h^{(i)}, y)$.

F^- substate	Role	Diagnostic from $h^{(i)}$ alone
Value conflict	speaker	Is my expression uniquely identifying <i>for me</i> ?
Uncertainty	speaker	Can I commit to a referent, or do candidates remain?
Contradiction	addressee	Does my partner's claim fit <i>any</i> candidate on my map?
Hazard	addressee	Are we acting confidently on a grounding I doubt?

No omniscient access required — this is answerable by the giver in the example slide, using only their own map.

Our fix predicts exactly where things break

EVIDENCE

Pending isn't one thing.

At the reference level → Uncertainty

- Identical: 70% routine existence checks (F^+ -like, resolves fast).
- **Multiplicity: 38% ungrounded** — follower can't commit to a candidate, exactly the Uncertainty diagnostic.

Implication for agents

Humans negotiate well in the moment. The bottleneck is *tracking* that a resolved ambiguity stays available to resurface — exactly what per-perspective friction is built to catch.

Alignment quietly decays.

Across the reference chain → Hazard

- **38.9%** of multiplicity chains reach **aligned**, then *revert* (vs. 9.9% identical) — confident, then silently wrong.
- Not from bad repair: silent accommodation is *rare* (0.2%). Decay comes from *new* references reopening an ambiguity already resolved once.

Now test it: can an LLM use perspective?

TEST

The question that decides everything: is detecting misalignment a *context-coverage* problem (more information $\Rightarrow$ better), or a *perspective* problem?

Omniscient

Both maps + prefix.

"Same physical landmark?"

Giver-only

Giver's map only.

"Underspecified on *your* map?"

Follower-only

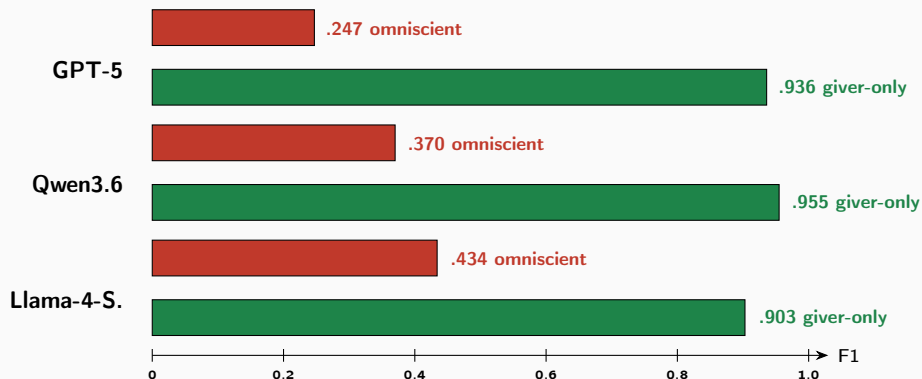
Follower's map only.

"Does this conflict with *your* map?"

200 REs (100 friction-warranted, 100 controls), 3 models: GPT-5, Qwen3.6-35B-A3B, Llama-4-Scout.

More information makes it *worse*

HERO RESULT



Consistent across closed/open, dense/MoE, full/quantized models — the gap is driven entirely by the friction subset, not the easy controls.

Why does more context hurt?

EXPLANATION

- Not a capacity issue: 35/100 friction items fail under *all three* models omnisciently; only 1 fails under all three with just the giver's map.
- Models' own rationales under omniscience *enumerate* the giver's two candidate instances — and then conclude alignment anyway.
- The omniscient prompt *elicits* the structural fact but doesn't force the model to *act* on it; the giver-only prompt makes ambiguity the question itself.

Takeaway

Restricting the model's view doesn't strip information — it removes the places to bury it.

Does the fix predict what actually gets repaired?

VALIDATION

Compose two single-perspective resolutions into a detector:

$$\text{Contr}(h^{(i)}, y) = 1 \left[\text{ref}(y \mid h^{(g)}) \neq \text{ref}(y \mid h^{(f)}) \right]$$

	Gemma-4-31B		GPT-5	
	Omni.	Persp.	Omni.	Persp.
Repairs (of 13)	5	13	12	13

A much *smaller* open model, perspective-bound, catches every repairable misalignment. Omniscient frontier access still misses one.

The signal isn't just more accurate — it tracks exactly what the dialogue goes on to fix.

Does this generalize beyond MapTask?

GENERALIZATION

OneCommon (Udagawa & Aizawa, 2019): continuous, not discrete, perceptual overlap — a much harder setting.

- 56% selection-failure rate (vs. 1.8% in MapTask) — but the same principle holds.
- *Quantity* of friction doesn't predict success: failed dialogues hedge *more* ($p < 10^{-19}$).
- What matters is whether friction is *productive* — the same F^+/F^- logic, a different corpus.

TAKEAWAY

Problem

Perceptual asymmetry breaks the shared-history assumption every current FPO system relies on — and it's concentrated in a specific, identifiable pattern.

Solution

Evaluate friction per-perspective. It requires *less* access, not more — and it's what makes detection work at all.

Impact

Perspective beats omniscience at every scale tested — a structural fix, not a bigger-model fix — and it predicts real repair.

Free annotation refinements: expose pending subtypes; distinguish negotiated from accommodated alignment.

Wherever agents coordinate under private information,
friction-aware systems must reason *from a perspective*,
not over all of them.

Future work: perspectival re-annotation of OneCommon; transfer to DPIP; ToM belief-state tracking to expose silent accommodation.

Thank you! Questions?

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